

The derivative-term conjecture for x^{x^2}

Forced cancellations, nonvanishing regions,
and certified computations for OEIS A290268

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Abstract

Let $a(n)$ be the number of nonzero monomials in the fully collected n th derivative of x^{x^2} . The conjecture in OEIS A290268 predicts

$$a(n) = U(n) := \frac{n^2}{2} + n + 1 - (n \bmod 2) \left(\frac{1}{2} + \left\lfloor \frac{n+1}{8} \right\rfloor \right).$$

This article does *not* establish the equality for all n . It proves the unconditional bound $a(n) \leq U(n)$ by identifying every cancellation needed to explain the proposed formula. It also proves positivity throughout the upper half of the coefficient triangle, classifies all zeros on the first two logarithmic-deficit diagonals, and gives explicit quadratic lower bounds. An exact generating-function identity and a finite-difference formula reduce the full conjecture to a precise nonvanishing assertion. Integer arithmetic verifies the support through $n = 200$; two complete modular computations, combined with the proved zero identities, certify $a(n) = U(n)$ for every $0 \leq n \leq 3000$. All source code, finite certificates, and reproduction instructions accompany the article. The remaining nonvanishing step is stated explicitly rather than assumed.

1 The problem and the status of the result

Put

$$f(x) = x^{x^2} = \exp(x^2 \log x), \quad x > 0.$$

The sequence A290268 counts the terms in the fully expanded derivative $f^{(n)}(x)$, after identical monomials have been collected. The OEIS entry was contributed by Vladimir Reshetnikov in October 2017; its formula section labels both the generating function and the explicit expression as conjectural. Peter Luschny's equivalent floor-function form is explicitly conditional on that conjecture [1].

For clarity, throughout this article the letter U denotes the *proposed* answer, not an already proved formula for a :

$$U(n) = \frac{n^2}{2} + n + 1 - (n \bmod 2) \left(\frac{1}{2} + \left\lfloor \frac{n+1}{8} \right\rfloor \right). \quad (1.1)$$

Equivalently,

$$U(2r) = 2r^2 + 2r + 1, \quad U(2r+1) = 2(r+1)^2 - \left\lfloor \frac{r+1}{4} \right\rfloor. \quad (1.2)$$

The two central issues are different. One must explain why certain coefficients vanish, and one must also exclude every other cancellation. The first task is completed here. The second is completed in substantial regions and in a large finite range, but not in full generality.

Theorem 1.1 (Unconditional bounds). *For every integer $n \geq 0$,*

$$\Lambda(n) \leq a(n) \leq U(n), \quad \Lambda(n) = \begin{cases} (3n^2 + 10n + 8)/8, & n \text{ even}, \\ (3n^2 + 12n + 1)/8, & n \text{ odd}. \end{cases} \quad (1.3)$$

In particular, $a(n) = \Theta(n^2)$. The proposed leading constant $1/2$ is not established by these bounds.

The upper bound is proved in Section 6; the lower bound is proved in Section 4. Section 9 gives a reproducible finite certification of the stronger equality through 3000. The exact outstanding assertion appears in Section 10.

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2 A canonical polynomial expansion

Set $z = x^2$ and $L = \log x$. Define polynomials $P_n \in \mathbb{Z}[z, L]$ by

$$f^{(n)}(x) = f(x)x^{-n}P_n(x^2, \log x), \quad P_0 = 1. \quad (2.1)$$

Write

$$P_n(z, L) = \sum_{k,j} c_{n,k,j} z^k L^j. \quad (2.2)$$

Thus $c_{n,k,j}$ is the coefficient of $x^{x^2+2k-n}(\log x)^j$ in the derivative. Counting terms means counting nonzero $c_{n,k,j}$, rather than counting the summands generated by a particular symbolic differentiation algorithm.

Lemma 2.1 (Uniqueness of the collected expansion). *A finite identity $\sum_{k,j} b_{k,j} x^{2k} (\log x)^j = 0$ for all $x > 0$ forces every $b_{k,j}$ to vanish. Consequently, the expansion (2.1)–(2.2) is canonical.*

Proof. Substitute $x = e^L$ and group the identity as $\sum_k e^{2kL} B_k(L) = 0$, with polynomial B_k . If K is the largest index with $B_K \neq 0$, division by e^{2KL} shows that $B_K(L)$ tends to zero as $L \rightarrow +\infty$: all remaining terms are polynomials multiplied by decaying exponentials. A nonzero polynomial cannot tend to zero at $+\infty$. This contradiction eliminates K , and then all indices. $\square$

The normalized derivative obeys a particularly small recurrence.

Proposition 2.2 (Polynomial and coefficient recurrences). *For every $n \geq 0$,*

$$P_{n+1} = (z(2L+1) - n)P_n + 2z \frac{\partial P_n}{\partial z} + \frac{\partial P_n}{\partial L}. \quad (2.3)$$

With nonexistent coefficients interpreted as zero,

$$\boxed{c_{n+1,k,j} = (2k - n)c_{n,k,j} + (j+1)c_{n,k,j+1} + c_{n,k-1,j} + 2c_{n,k-1,j-1}.} \quad (2.4)$$

For $n \geq 1$, the only possible nonzero indices satisfy

$$1 \leq k \leq n, \quad 0 \leq j \leq k. \quad (2.5)$$

Proof. We have $f'/f = x(2L+1)$, $dz/dx = 2z/x$, and $dL/dx = 1/x$. Differentiate (2.1) and factor out $f(x)x^{-n-1}$ to obtain (2.3). Taking the coefficient of $z^k L^j$ gives (2.4). Starting from $P_0 = 1$, the first derivative is $P_1 = z(2L+1)$. The four operations in (2.3) preserve $k \geq 1$ and $j \leq k$, and increase k by at most one. Induction proves (2.5) and integrality. $\square$

For example,

$$\begin{aligned} P_1 &= z(2L+1), \\ P_2 &= z(2L+3) + z^2(2L+1)^2, \\ P_3 &= 2z + z^2(12L^2 + 24L + 9) + z^3(2L+1)^3. \end{aligned}$$

These have respectively 2, 5, and 8 nonzero coefficients, in agreement with the definition. Already P_3 illustrates a degree loss: its z -coefficient has no L -term. Later cancellations are less obvious. For instance,

$$[z^2]P_7(z, L) = 28, \quad [z^2]P_9(z, L) = -864L. \quad (2.6)$$

The missing L -term in the first polynomial and the missing constant in the second will follow from the same reflection principle.

3 Exact formulas for individual coefficients

3.1 An exponential generating function

Define two formal power series

$$A(t) = t(2+t) = (1+t)^2 - 1, \quad B(t) = (1+t)^2 \log(1+t). \quad (3.1)$$

Formal coefficient extraction is denoted by $[t^n]$. All logarithms in formal series are expanded at the origin.

Proposition 3.1 (Generating function). *The complete coefficient array is described by*

$$\sum_{n \geq 0} P_n(z, L) \frac{t^n}{n!} = \exp(z(A(t)L + B(t))). \quad (3.2)$$

In particular, if $m = k - j \geq 0$, then

$$c_{n,k,j} = \frac{n!}{j!m!} [t^n] A(t)^j B(t)^m. \quad (3.3)$$

Proof. Taylor expansion at the point x gives

$$\frac{f(x(1+t))}{f(x)} = \sum_{n \geq 0} \frac{x^n f^{(n)}(x)}{f(x)} \frac{t^n}{n!}.$$

The logarithm of the ratio on the left is

$$x^2(((1+t)^2 - 1) \log x + (1+t)^2 \log(1+t)).$$

This proves (3.2) after substitution of z, L . Alternatively, its coefficients satisfy (2.3), so the identity also holds with z, L algebraically independent. Expanding separately the two exponentials $\exp(zA(t)L)$ and $\exp(zB(t))$ gives (3.3). $\square$

Because both A and B start with a nonzero term of degree one, their product in (3.3) starts at degree k . This independently explains $k \leq n$. The difficulty is that B does not have only positive coefficients:

$$B(t) = t + \frac{3}{2}t^2 + \frac{1}{3}t^3 - \frac{1}{12}t^4 + \frac{1}{30}t^5 - \dots$$

Consequently, the coefficient formula is not itself a proof of nonvanishing.

3.2 Falling factorials and finite differences

Let

$$F_n(u) = (u)_{\underline{n}} = u(u-1) \cdots (u-n+1), \quad F_0(u) = 1,$$

and let $\Delta_2 G(u) = G(u+2) - G(u)$. Set $Q_{n,j}(u) = \Delta_2^j F_n(u)$.

Proposition 3.2 (Finite-difference formula). *For $m = k - j \geq 0$,*

$$c_{n,k,j} = \frac{1}{j!m!} \left. \frac{d^m}{du^m} Q_{n,j}(u) \right|_{u=2m} \quad (3.4)$$

$$= \frac{1}{j!} [v^m] \sum_{h=0}^j (-1)^{j-h} \binom{j}{h} F_n(2m+2h+v). \quad (3.5)$$

Proof. The generalized binomial theorem gives $n![t^n](1+t)^u = F_n(u)$. Differentiating m times in u introduces $\log^m(1+t)$. Taking j forward differences of step two introduces $((1+t)^2 - 1)^j = A(t)^j$. Evaluation at $u = 2m$ introduces $(1+t)^{2m}$, which combines with the logarithm to give $B(t)^m$. Equation (3.3) now proves (3.4); Taylor's formula proves (3.5). $\square$

Derivatives of falling factorials also occur in the theory of shifted Stirling numbers and elementary symmetric harmonic sums; Poon gives a broader account of these connections [2]. No general nonvanishing theorem from that literature is being assumed here. Formula (3.5) is also an independent way to compute a coefficient using integer polynomial products.

4 A transformation proving positivity

A fractional-linear change of variable turns a large region of the coefficient array into a manifestly positive expression. Define

$$H(u) = \frac{\operatorname{atanh} u}{u} = \sum_{r \geq 0} \frac{u^{2r}}{2r+1}, \quad H(0) = 1. \quad (4.1)$$

The value at zero is understood by its formal power series.

Proposition 4.1 (Transformed coefficient formula). *For $m = k - j \geq 0$ and $n \geq k$,*

$$c_{n,k,j} = \frac{n!}{j!m!} 2^{2j+m-n} [u^{n-k}](1-u)^{n-2k-1}(1+u)^{2m} H(u)^m. \quad (4.2)$$

Negative integer powers of $1-u$ have their usual formal expansions.

Proof. In the residue expression for $[t^n]A(t)^j B(t)^m$, substitute $t = 2u/(1-u)$. Then

$$1+t = \frac{1+u}{1-u}, \quad A(t) = \frac{4u}{(1-u)^2}, \quad B(t) = \frac{2(1+u)^2}{(1-u)^2} \operatorname{atanh} u, \quad dt = \frac{2 du}{(1-u)^2}.$$

Using $[t^n]G(t) = \operatorname{Res}_{t=0} G(t)t^{-n-1} dt$, the constant factor becomes 2^{2j+m-n} . The remaining power of $1-u$ is $n-2k-1$. The factors $u^j(\operatorname{atanh} u)^m$ contribute $u^k H(u)^m$, so the resulting residue extracts the coefficient of u^{n-k} . Multiplying by $n!/(j!m!)$ proves the formula. The formal change-of-variable rule applies because $2u/(1-u)$ has zero constant term and nonzero linear coefficient. $\square$

Theorem 4.2 (A nonvanishing region). *For $1 \leq k \leq n$ and $0 \leq j \leq k$,*

$$\begin{aligned} n \leq 2k &\implies c_{n,k,j} > 0, \\ n = 2k+1, j < k &\implies c_{n,k,j} > 0. \end{aligned} \quad (4.3)$$

Proof. If $n \leq 2k$, the factor $(1-u)^{n-2k-1}$ is $(1-u)^{-q}$ with $q \geq 1$. Every coefficient of this factor is strictly positive. The other two factors in (4.2) have nonnegative coefficients and constant term one. Their product therefore has a strictly positive coefficient in every nonnegative degree.

If $n = 2k+1$, that factor disappears. The condition $j < k$ means $m \geq 1$. The series $H(u)^m$ has a strictly positive coefficient in every even degree. The polynomial $(1+u)^{2m}$ has positive constant and linear coefficients. An even target degree can be supplied entirely by $H(u)^m$; an odd target degree can use the linear term of $(1+u)^{2m}$ and an even degree from $H(u)^m$. Thus the target coefficient of degree $n-k$ is strictly positive in both cases. The prefactor in (4.2) is positive. $\square$

Corollary 4.3 (Quadratic lower bounds). *The lower bound in Theorem 1.1 holds.*

Proof. For $n = 2r \geq 2$, Theorem 4.2 supplies all coefficients with $r \leq k \leq 2r$, giving

$$a(2r) \geq \sum_{k=r}^{2r} (k+1) = \frac{(r+1)(3r+2)}{2}.$$

For $n = 2r + 1$ with $r \geq 1$, it supplies all coefficients with $r + 1 \leq k \leq 2r + 1$ and the r coefficients $0 \leq j < r$ at $k = r$. Hence

$$a(2r+1) \geq r + \sum_{k=r+1}^{2r+1} (k+1) = \frac{3(r+1)(r+2)}{2} - 1.$$

These are precisely (1.3). The cases $n = 0, 1$ follow directly from $P_0 = 1$ and $P_1 = z(2L + 1)$. $\square$

5 The two forced families of zeros

5.1 Degree-forced zeros on the top diagonal

Setting $j = k$, and therefore $m = 0$, in (3.3) yields an explicit coefficient.

Proposition 5.1 (Top logarithmic degree). *For $k \geq 1$ and $k \leq n \leq 2k$,*

$$c_{n,k,k} = \frac{n!}{k!} \binom{k}{n-k} 2^{2k-n} > 0. \quad (5.1)$$

For $n > 2k$, one has $c_{n,k,k} = 0$.

Proof. The relevant series is the polynomial $A(t)^k = t^k(2+t)^k$, of degree $2k$. Its coefficient of t^n is $\binom{k}{n-k} 2^{2k-n}$ in the indicated range and zero above it. $\square$

For a fixed $n \geq 1$, this removes exactly $\lfloor (n-1)/2 \rfloor$ cells from the possible triangle.

5.2 Reflection-forced cancellations

The less obvious zeros follow from symmetry about the midpoint of the roots of a falling factorial.

Lemma 5.2 (Reflection of a finite difference). *For all $n, j \geq 0$,*

$$Q_{n,j}(n-2j-1-u) = (-1)^{n-j} Q_{n,j}(u). \quad (5.2)$$

Proof. First, $F_n(n-1-u) = (-1)^n F_n(u)$, by reversing the order of the factors. Expand the left side of (5.2):

$$\sum_{h=0}^j (-1)^{j-h} \binom{j}{h} F_n(n-2j-1-u+2h).$$

Reflection turns its last factor into $(-1)^n F_n(u+2(j-h))$. Reindexing by $\ell = j-h$ leaves the factor $(-1)^{n-j}$ times the usual expansion of $\Delta_2^j F_n(u)$. $\square$

Theorem 5.3 (Reflection zeros). *Suppose $1 \leq k \leq n$ and $0 \leq j < k$. Then*

$$k \text{ even}, \quad n = 4k - 2j + 1 \quad \implies \quad c_{n,k,j} = 0. \quad (5.3)$$

Proof. Put $m = k - j$. The reflection center in Lemma 5.2 is $(n - 2j - 1)/2$. Under the displayed condition this equals $2m$, precisely the evaluation point in (3.4). The parity of $Q_{n,j}(2m + v)$ as a polynomial in v is $n - j$. Its coefficient of v^m therefore vanishes whenever $n - j - m = n - k$ is odd. The relation $n = 4k - 2j + 1$ makes n odd, so $n - k$ is odd when k is even. Equation (3.5) completes the proof. $\square$

There is also a direct proof using (4.2). The relation in (5.3) gives $n - 2k - 1 = 2m$, so the series whose coefficient is extracted becomes

$$(1 - u^2)^{2m} H(u)^m. \quad (5.4)$$

This series is even. When k is even, $n - k$ is odd, and its coefficient of that degree is zero. This second proof makes the cancellation visible without evaluating any large integer coefficient.

Remark 5.4 (What symmetry does not say). The theorem is an implication, not an equivalence. An even power series has zero odd coefficients, but it can also have zero even coefficients. Similarly, away from the reflection center, a derivative of a polynomial can vanish for reasons unrelated to symmetry. Excluding such additional zeros is a separate requirement of the full conjecture.

6 Counting the forced zeros

For $n \geq 1$, the triangle (2.5) has

$$T(n) = \sum_{k=1}^n (k+1) = \frac{n(n+3)}{2} \quad (6.1)$$

cells. Let Z_n^{top} be the top-diagonal cells removed by Proposition 5.1, and let Z_n^{ref} be the cells in Theorem 5.3. These sets are disjoint, since the second requires $j < k$.

Lemma 6.1 (Number of reflection zeros). *For even n , Z_n^{ref} is empty. For odd n ,*

$$|Z_n^{\text{ref}}| = \left\lfloor \frac{n+1}{8} \right\rfloor. \quad (6.2)$$

Proof. Reflection zeros have odd n . Write $n = 2r + 1$ and $k = 2\ell$. The reflection relation determines $j = 2k - r = 4\ell - r$. The inequalities $0 \leq j < k$ become

$$\frac{r}{4} \leq \ell < \frac{r}{2}.$$

For $r \geq 1$, these automatically give $\ell \geq 1$ whenever the interval contains an integer. Its number of integers is

$$\left\lceil \frac{r}{2} \right\rceil - \left\lceil \frac{r}{4} \right\rceil = \left\lfloor \frac{r+1}{4} \right\rfloor,$$

as is checked in the four residue classes of r modulo four. For $r = 0$ both sides are zero. Since $(r+1)/4 = (n+1)/8$, the result follows. $\square$

Theorem 6.2 (The conjectured expression is an upper bound). *For every $n \geq 0$,*

$$a(n) \leq U(n). \quad (6.3)$$

Equality holds for a given n if and only if the coefficient triangle has no zeros other than $Z_n^{\text{top}} \cup Z_n^{\text{ref}}$.

Proof. For even $n = 2r \geq 2$, subtraction of the top-diagonal zeros from (6.1) gives

$$2r^2 + 3r - (r - 1) = 2r^2 + 2r + 1.$$

For odd $n = 2r + 1$, subtraction of both disjoint families gives

$$2r^2 + 5r + 2 - r - \left\lfloor \frac{r+1}{4} \right\rfloor = 2(r+1)^2 - \left\lfloor \frac{r+1}{4} \right\rfloor.$$

These are (1.2). Additional zeros could only decrease the number of nonzero coefficients, proving the inequality and the stated equivalence. Finally $a(0) = U(0) = 1$. $\square$

For example, at $n = 7$ there are 35 potential cells, three top-diagonal zeros, and the reflection zero $(k, j) = (2, 1)$, leaving 31. At $n = 9$, there are 54 potential cells, four top-diagonal zeros, and the reflection zero $(2, 0)$, leaving 49. The exact expansions in (2.6) display those two different cancellations. The floor term in the proposed formula is thus explained by a proved symmetry, not merely by fitting initial values.

7 Nonvanishing on the first two deficit diagonals

Call $m = k - j$ the *logarithmic deficit*. This section gives a complete zero classification for $m = 1$ and $m = 2$. Together with Theorem 4.2, it narrows the unproved portion of the problem.

7.1 A beta-integral representation

Suppose $n > 2k$, $m = k - j \geq 1$, and put

$$d = n - 2k - 1 \geq 0. \tag{7.1}$$

For sufficiently small complex ε , define

$$I_{m,j,d}(\varepsilon) = \int_0^1 t^{2m+\varepsilon} (1-t)^{d-\varepsilon} (2t-1)^j dt. \tag{7.2}$$

The integral and all derivatives required below converge uniformly on compact subsets of $|\varepsilon| < 1/2$.

Lemma 7.1 (Integral formula). *Under these assumptions,*

$$Q_{n,j}(2m + \varepsilon) = (-1)^{n-1} n! \frac{\sin(\pi\varepsilon)}{\pi} I_{m,j,d}(\varepsilon), \tag{7.3}$$

and consequently

$$c_{n,k,j} = \frac{(-1)^{n-1} n!}{j!} [\varepsilon^m] \left(\frac{\sin(\pi\varepsilon)}{\pi} I_{m,j,d}(\varepsilon) \right). \tag{7.4}$$

Proof. Euler's reflection formula and beta integral [3, 4] give, initially away from integer α ,

$$\begin{aligned} F_n(\alpha) &= (-1)^{n-1} \frac{\sin(\pi\alpha)}{\pi} \Gamma(\alpha+1) \Gamma(n-\alpha) \\ &= (-1)^{n-1} n! \frac{\sin(\pi\alpha)}{\pi} \int_0^1 t^\alpha (1-t)^{n-\alpha-1} dt. \end{aligned}$$

In the j th difference put $\alpha = 2m + 2h + \varepsilon$. The sine factor is independent of the integer h . The finite sum inside the integral is

$$\sum_{h=0}^j (-1)^{j-h} \binom{j}{h} \left(\frac{t}{1-t} \right)^{2h} = \frac{(2t-1)^j}{(1-t)^{2j}}.$$

The remaining exponent of $1-t$ is $d-\varepsilon$, proving (7.3). All terms are analytic near $\varepsilon = 0$, so the identity extends to zero as well. Taking the coefficient of ε^m and using (3.5) gives (7.4). $\square$

7.2 Deficit one

Theorem 7.2 (Complete signs for $m = 1$). *Suppose $m = k - j = 1$ and $n > 2k$, and define d by (7.1). Then*

$$\operatorname{sgn} c_{n,k,j} = (-1)^{n-1} \begin{cases} 1, & j \text{ even}, \\ \operatorname{sgn}(2-d), & j \text{ odd}. \end{cases} \quad (7.5)$$

In particular the only zeros in this range have j odd and $d = 2$.

Proof. The coefficient of ε in (7.4) is

$$c_{n,k,j} = \frac{(-1)^{n-1} n!}{j!} \int_0^1 t^2 (1-t)^d (2t-1)^j dt. \quad (7.6)$$

For even j , the integrand is nonnegative and positive except at finitely many points, so the integral is strictly positive. For odd j , pair t with $1-t$ and integrate over $1/2 < t < 1$. The paired integrand is

$$(2t-1)^j (t^2(1-t)^d - t^d(1-t)^2).$$

The ratio of the two positive terms in parentheses is $(t/(1-t))^{2-d}$. Since $t/(1-t) > 1$, their difference is identically zero when $d = 2$ and otherwise has the strict sign of $2-d$ throughout the open interval. This proves the sign formula. $\square$

These zeros are exactly the previously identified reflection zeros: j odd means $k = j + 1$ is even, and $d = 2$ means $n = 2k + 3 = 4k - 2j + 1$.

7.3 Deficit two

Theorem 7.3 (Complete signs for $m = 2$). *Suppose $m = k - j = 2$ and $n > 2k$. Then*

$$\operatorname{sgn} c_{n,k,j} = (-1)^{n-1} \begin{cases} 1, & j \text{ odd}, \\ \operatorname{sgn}(4-d), & j \text{ even}. \end{cases} \quad (7.7)$$

The only zeros in this range have j even and $d = 4$.

Proof. The coefficient of ε^2 in (7.4) is

$$c_{n,k,j} = \frac{(-1)^{n-1} n!}{j!} \int_0^1 t^4 (1-t)^d (2t-1)^j \log \frac{t}{1-t} dt. \quad (7.8)$$

For odd j , the factors $(2t-1)^j$ and $\log(t/(1-t))$ have the same sign, so the integral is strictly positive. For even j , their product is antisymmetric under $t \mapsto 1-t$. Pairing the integral as before leaves, on $1/2 < t < 1$,

$$(2t-1)^j \log \frac{t}{1-t} (t^4(1-t)^d - t^d(1-t)^4).$$

The first two factors are positive there. The bracket has the strict sign of $4 - d$, or is identically zero when $d = 4$. This proves the claim. $\square$

Again the zero condition is exactly reflection: $k = j + 2$ is even and $n = 2k + 5 = 4k - 2j + 1$.

For $n \leq 2k$, positivity has already been proved, and the borderline $n = 2k + 1$ is included in both sign arguments. Thus these two theorems, together with Proposition 5.1, classify all zeros for $k - j \leq 2$ throughout the possible triangle.

7.4 Why the same sign proof does not settle higher deficits

At $m = 3$, formula (7.4) gives the integral of

$$t^6(1-t)^d(2t-1)^j \left(\frac{1}{2} \log^2 \frac{t}{1-t} - \frac{\pi^2}{6} \right) \quad (7.9)$$

multiplied by $(-1)^{n-1}n!/j!$. The last factor changes sign inside $(0, 1)$. Thus positivity of the weight, or pairing t with $1 - t$, no longer excludes cancellation. This is a concrete obstruction to simply extending the proofs above. It is not a counterexample to the conjecture.

8 Generating functions and equivalent formulas for the upper bound

All identities in this section are proved identities for $U(n)$. They become identities for $a(n)$ only after the remaining nonvanishing assertion is supplied, or when restricted to a certified finite range.

8.1 Derivation of the rational generating function

By (1.2), the even and odd subsequences have generating functions

$$E(y) = \sum_{r \geq 0} U(2r)y^r = \frac{(1+y)^2}{(1-y)^3}, \quad (8.1)$$

$$O(y) = \sum_{r \geq 0} U(2r+1)y^r = \frac{2(1+y)}{(1-y)^3} - \frac{y^3}{(1-y)(1-y^4)}. \quad (8.2)$$

For the last term, write $\lfloor (r+1)/4 \rfloor = \sum_{h \geq 1} \mathbf{1}_{r \geq 4h-1}$ and sum each geometric tail. Therefore

$$\begin{aligned} G(t) &:= \sum_{n \geq 0} U(n)t^n = E(t^2) + tO(t^2) \\ &= \frac{1+t^2}{(1-t)^3(1+t)} - \frac{t^7}{(1-t^2)(1-t^8)} \\ &= \frac{N(t)}{(1-t)(1-t^2)(1-t^8)}, \end{aligned} \quad (8.3)$$

where

$$N(t) = 1 + t + 2t^2 + 2t^3 + 2t^4 + 2t^5 + 2t^6 + t^7 + 2t^8 + t^9. \quad (8.4)$$

Indeed, $N(t) = (1+t^2)(1+t+\dots+t^7) - t^7(1-t)$. This is exactly the rational function proposed in the OEIS entry [1]. The derivation shows why period eight appears: it counts the even values of k on the reflection line, in addition to the parity split of the ordinary triangle.

8.2 The eight constituent polynomials

Writing $n = 8q + s$ with $0 \leq s \leq 7$, (1.1) becomes the following degree-two quasipolynomial.

s	$U(8q + s)$
0	$32q^2 + 8q + 1$
1	$32q^2 + 15q + 2$
2	$32q^2 + 24q + 5$
3	$32q^2 + 31q + 8$
4	$32q^2 + 40q + 13$
5	$32q^2 + 47q + 18$
6	$32q^2 + 56q + 25$
7	$32q^2 + 63q + 31$

The trigonometric expression from the OEIS entry is another way to write the same eight polynomials:

$$U(n) = \frac{8n^2 + 15n + 14 + (n+2)(-1)^n + (2 - 4\sqrt{2}\sin(\pi n/4))\sin(\pi n/2)}{16}. \quad (8.5)$$

To verify the equivalence without an approximation, substitute $n = 8q + s$. The sine factors take their exact values at multiples of $\pi/4$ and $\pi/2$. Substitution in each of the eight residue classes yields the polynomials in the table. In particular, no irrational value remains.

8.3 Linear recurrences

Multiplication of (8.3) by its denominator yields

$$\begin{aligned} U(n) = & U(n-1) + U(n-2) - U(n-3) + U(n-8) \\ & - U(n-9) - U(n-10) + U(n-11), \quad n \geq 11. \end{aligned} \quad (8.6)$$

Multiplying the denominator by $1+t$ gives the convenient order-twelve form recorded in the OEIS entry:

$$U(n) = 2U(n-2) - U(n-4) + U(n-8) - 2U(n-10) + U(n-12), \quad n \geq 12. \quad (8.7)$$

The twelve initial values are

$$1, 2, 5, 8, 13, 18, 25, 31, 41, 49, 61, 71.$$

Using these recurrences to *generate* the claimed values of $a(n)$ would be circular in a proof of the conjecture. The computations in the next section instead use the independently derived coefficient recurrence (2.4).

9 Exact finite verification and certificates

9.1 Integer arithmetic and independent formula checks

The accompanying Python program applies (2.4) with arbitrary-precision integers. Through $n = 200$, it checks not only the term count but the entire zero pattern: a coefficient is zero if and only if it belongs to one of the two proved zero families. This examines 1,373,500 cells with $1 \leq k \leq n \leq 200$, $0 \leq j \leq k$.

For $0 \leq n \leq 16$, the program independently recomputes 969 triples (n, k, j) , including the $k = 0$ boundary, using each of (3.3), (3.5), and (4.2). Fractions are exact rational numbers. All three formulas agree with the integer recurrence. The positivity and deficit-one and -deficit-two sign theorems are also checked throughout the integer range. These checks test the derivations and their implementation; the proofs in the preceding sections do not depend on them.

9.2 Why modular arithmetic gives a rigorous finite certificate

A nonzero residue modulo any integer $M \geq 2$ proves that the original integer is nonzero. A zero residue, in contrast, might mean only that M divides a nonzero integer. We use two moduli,

$$M_1 = 1,000,000,007, \quad M_2 = 1,000,000,009. \quad (9.1)$$

Their primality is not needed for this argument: only exact reduction of an integer recurrence is used, with no modular division.

For each modulus, a C++ program computes every row through $n = 3000$ and records every zero residue outside the two mathematically proved zero families. There are three such residues modulo M_1 and four modulo M_2 . The two lists are disjoint. Their union and the corresponding residues are as follows.

n	k	j	$c_{n,k,j} \bmod M_1$	$c_{n,k,j} \bmod M_2$
1471	1046	521	847578491	0
1701	610	505	0	439228697
1854	1713	549	0	366891231
2336	1689	145	0	782191592
2543	1785	663	405694396	0
2807	2566	502	607368440	0
2842	2673	2118	474540977	0

Every entry in this table is therefore a nonzero integer coefficient. All other cells outside the proved zero families have a nonzero residue in both runs. Each modulus run checks 4,509,002,501 potential cells, including the single cell at $n = 0$.

Proposition 9.1 (Finite computational certification). *The accompanying exact computations, together with Proposition 5.1 and Theorem 5.3, certify*

$$a(n) = U(n) \quad \text{for every integer } 0 \leq n \leq 3000. \quad (9.2)$$

Certificate argument. The two analytic results prove that every predicted zero is an actual integer zero. For any other coefficient in the finite range, membership in neither, one, or both modular zero lists exhausts the possibilities. The intersection of the lists is empty. Thus at least one of its two residues is nonzero, proving that the integer coefficient is nonzero. Theorem 6.2 now gives the equality for every row in the range. $\square$

This is a deterministic finite verification, not a probabilistic claim based on the size of the moduli. It is also not a formal proof-assistant certificate: it relies on the supplied ordinary programs and their execution, which can be independently reproduced.

9.3 Implementation details and limitations

The C++ program stores two coefficient rows in arrays of unsigned 32-bit integers. An update is evaluated in signed 64-bit arithmetic before reduction. Its explicit input bounds guarantee that these intermediates cannot overflow. It never approximates logarithms or transcendental values. The arrays have $O(N^2)$ entries and the computation uses $O(N^3)$ fixed-modulus arithmetic operations through order N .

The Python implementation uses a different update organization: it scatters each existing coefficient into its four successors, whereas the C++ implementation gathers four contributions into each destination. The small-range generating-function and finite-difference calculations provide further independent comparisons. The archive retains both raw modular zero lists, row-by-row support statistics, explicit nonzero witnesses, the combined certificate, and the exact small-range counts.

No finite range, regardless of its size, proves the nonvanishing assertion for all n . In particular, the certificate supplies neither an induction step beyond 3000 nor a theorem bounding where a first additional zero could occur.

10 The precise missing step

The coefficient problem can now be stated independently of the original transcendental function.

Conjecture 10.1 (Complete support classification). Let $n \geq 1$, $1 \leq k \leq n$, $0 \leq j \leq k$, and $m = k - j$. Then

$$\frac{1}{j!m!} \frac{d^m}{du^m} \Delta_2^j(u) \Big|_{u=2m} = 0 \quad (10.1)$$

if and only if at least one of the following holds:

$$(j = k \text{ and } n > 2k), \quad (k \text{ even and } n = 4k - 2j + 1). \quad (10.2)$$

The implication from (10.2) to (10.1) is proved above. The converse remains unproved in the following precise region:

$$m = k - j \geq 3, \quad n \geq 2k + 2, \quad (10.3)$$

away from the reflection zeros. All other regions have been handled by the preceding analytic theorems.

Theorem 10.2 (Exact reduction of the OEIS conjecture). *The equality $a(n) = U(n)$ for every $n \geq 0$ is equivalent to Conjecture 10.1. It is also equivalent to proving the nonvanishing assertion only in the remaining region (10.3), excluding the reflection-zero family.*

Proof. Formula (3.4) identifies the expression in (10.1) with the actual derivative coefficient. Theorem 6.2 counts exactly the cells left after removal of the two proved, disjoint zero families. Hence any additional zero makes the corresponding inequality strict, and no additional zeros makes it an equality. Theorem 4.2, Proposition 5.1, and Theorems 7.2–7.3 eliminate the complementary regions. $\square$

A tempting shortcut would assert that a derivative of a falling factorial cannot vanish at an integer other than its reflection center. That assertion is false. Direct differentiation gives

$$\frac{d^5}{du^5}(u)_{\underline{8}} = 3360(u - 5)(u - 2)(2u - 7). \quad (10.4)$$

It vanishes at $u = 2$ and $u = 5$, as well as at the reflection center $u = 7/2$. This is *not* a counterexample to Conjecture 10.1, because the derivative order, evaluation point, and finite-difference order in that conjecture are linked in a special way. It does show why those constraints must be used in a genuine proof of the missing implication.

The transformed formula provides an alternative target. With $d = n - 2k - 1 \geq 1$, the outstanding task is to prove that

$$[u^{k+d+1}](1-u)^d(1+u)^{2m} \left(\frac{\operatorname{atanh} u}{u} \right)^m \quad (10.5)$$

is nonzero for $m \geq 3$, except when $d = 2m$ and k is even. The reflection case is transparent because the series then becomes even; the remaining finite-difference cancellations are the essential issue. Equation (10.5) is an equivalent residual lemma, not an additional theorem claimed by this article.

11 Conclusion

The proposed formula has a rigorous structural explanation. The obvious degree loss removes the top logarithmic coefficients with $n > 2k$. A separate reflection symmetry removes exactly $\lfloor (n+1)/8 \rfloor$ further coefficients for odd n . Their total produces the OEIS floor expression and its rational generating function as an unconditional upper bound.

A fractional-linear transformation proves that a large region of the coefficient triangle is strictly positive, and beta-integral arguments exclude additional zeros on the first two deficit diagonals. These results prove quadratic growth and reduce the full formula to the explicit residual nonvanishing problem (10.5). Exact finite computations settle every derivative order through 3000. The global equality is not proved here: its remaining obligation is the nonvanishing statement in (10.3), not any of the counting or generating-function manipulations.

A Reproducing the computations

The archive needs Python 3.10 or later and a C++17 compiler for the computations. Python uses only its standard library. The PDF can be rebuilt with a conventional LaTeX installation containing the packages listed in the source preamble.

Run the following commands from the archive directory. The Makefile wraps the same operations.

```
python3 code/verify_exact.py --max-n 200 --formula-n 16
mkdir -p build
g++ -O3 -std=c++17 -Wall -Wextra \
    code/verify_modular.cpp -o build/verify_modular
build/verify_modular 3000 1000000007 data/p1000000007 \
    data/watch_coordinates.csv
build/verify_modular 3000 1000000009 data/p1000000009 \
    data/watch_coordinates.csv
python3 code/combine_certificates.py
pdflatex -interaction=nonstopmode -halt-on-error \
    A290268_partial_results.tex
pdflatex -interaction=nonstopmode -halt-on-error \
    A290268_partial_results.tex
```

The watch file requests explicit residues at the seven exceptional coordinates found in the modular runs. It does not restrict the search: the C++ program checks every coefficient in every row

regardless of the watch file. The combiner checks that the complete exceptional zero lists are disjoint and that all listed witnesses are consistent with them.

For a smaller test, the optional watch file can be omitted. For example, `build/verify_modular 100 1000000007 data/test` performs the full modular recurrence through 100. A watch file containing indices beyond the chosen bound is rejected, rather than silently ignored.

B Files included in the archive

File or directory	Purpose
A290268_partial_results.tex	Complete article source.
A290268_partial_results.pdf	Compiled article.
README.md, Makefile	Status, dependency notes, and reproduction commands.
code/verify_exact.py	Arbitrary-precision recurrence and three independent coefficient-formula checks.
code/verify_modular.cpp	Complete modular coefficient search with explicit overflow bounds.
code/combine_certificates.py	Verification and combination of the two finite modular certificates.
data/exact_counts.csv	Exact term counts through 200.
data/certified_counts.csv	Certified term counts through 3000.
data/p1000000007_* and data/p1000000009_*	Complete exceptional zero lists, row statistics, witness residues, run logs, and summaries.
data/modular_witnesses.csv	The seven coordinates at which one residue vanishes, with the nonzero residue in the other modulus.
data/*summary.json	Machine-readable verification scope and results.
data/sample_coefficients.json	Fully collected low-order coefficient arrays and cancellation examples.
SHA256SUMS.txt	Integrity checks for the distributed files; these are not mathematical proof certificates.

References

- [1] The OEIS Foundation Inc., *A290268: Number of terms in the fully expanded n -th derivative of x^{x^2}* , contributed by Vladimir Reshetnikov, with formula comments by Peter Luschny, October 2017. Accessed September 20, 2026. <https://oeis.org/A290268>.
- [2] Steven S. Poon, *Higher Derivatives of the Falling Factorial and Related Generalizations of the Stirling and Harmonic Numbers*, arXiv:1401.2737, 2014. <https://arxiv.org/abs/1401.2737>.
- [3] NIST Digital Library of Mathematical Functions, §5.5(ii), especially Eq. 5.5.3, Euler’s reflection formula. Accessed September 20, 2026. <https://dlmf.nist.gov/5.5.E3>.
- [4] NIST Digital Library of Mathematical Functions, §5.12, especially Eq. 5.12.1, Euler’s beta integral. Accessed September 20, 2026. <https://dlmf.nist.gov/5.12.E1>.